Quickstart Guide: zzt2f (Typst Backend)

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This quickstart covers `zzt2f()`, the Typst-based alternative to `t2f()`. Where `t2f()` requires a full LaTeX distribution (pdflatex + pdfcrop + ImageMagick), `zzt2f()` needs only two lightweight dependencies: the tinytable R package and the Typst CLI (~40 MB binary). Typst auto-sizes pages, so no cropping step is needed.

Both functions share the same theme system. The key difference is the compilation backend: LaTeX vs Typst.

Installation

```r
# R package
install.packages("tinytable")

# Typst CLI
# macOS:   brew install typst
# Linux:   curl -fsSL https://typst.community | sh
# Windows: winget install --id Typst.Typst
```

```r
library(zztab2fig)
```

Verify that both dependencies are available:

```r
check_typst_deps()
#> Checking Typst dependencies for zzt2f()...
#> tinytable: OK
#> typst CLI: OK
#>   Version: typst 0.14.2 (unknown hash)
#>
#> All dependencies available. zzt2f() is ready to use.
```

Basic Usage

Convert a data frame to a table image:

```r
zzt2f(
  mtcars[1:6, 1:4],
  filename = "basic_table",
  sub_dir = output_dir,
  format = "png"
)
```

	mpg	cyl	disp	hp
Mazda RX4	21.0	6	160	110
Mazda RX4 Wag	21.0	6	160	110
Datsun 710	22.8	4	108	93
Hornet 4 Drive	21.4	6	258	110
Hornet Sportabout	18.7	8	360	175
Valiant	18.1	6	225	105

By default, `zzt2f()` produces a PDF. Here we use `format = "png"` for display. The output is a single file with no intermediate artifacts (no `.tex`, no `_full.pdf`, no cropping step).

Output Formats

`zzt2f()` supports three output formats via the `format` argument:

```
zzt2f(mtcars[1:6, 1:4],
      filename = "fmt_pdf",
      sub_dir = output_dir,
      format = "pdf")
```

```
zzt2f(mtcars[1:6, 1:4],
      filename = "fmt_png",
      sub_dir = output_dir,
      format = "png",
      dpi = 300)
```

	mpg	cyl	disp	hp
Mazda RX4	21.0	6	160	110
Mazda RX4 Wag	21.0	6	160	110
Datsun 710	22.8	4	108	93
Hornet 4 Drive	21.4	6	258	110
Hornet Sportabout	18.7	8	360	175
Valiant	18.1	6	225	105

```
zzt2f(mtcars[1:6, 1:4],
      filename = "fmt_svg",
      sub_dir = output_dir,
      format = "svg")
```

PNG output respects the `dpi` argument (default 300). SVG output produces resolution-independent vector graphics suitable for web documents.

Journal Themes

`zzt2f()` reuses the same theme system as `t2f()`. All built-in themes translate automatically to Typst styling:

```
zzt2f(mtcars[1:6, 1:4],
      theme = "nejm",
```

```
      filename = "nejm_table",
      sub_dir = output_dir,
      format = "png")
```

	mpg	cyl	disp	hp
Mazda RX4	21.0	6	160	110
Mazda RX4 Wag	21.0	6	160	110
Datsun 710	22.8	4	108	93
Hornet 4 Drive	21.4	6	258	110
Hornet Sportabout	18.7	8	360	175
Valiant	18.1	6	225	105

```
zzt2f(mtcars[1:6, 1:4],
      theme = "apa",
      filename = "apa_table",
      sub_dir = output_dir,
      format = "png")
```

	mpg	cyl	disp	hp
Mazda RX4	21.0	6	160	110
Mazda RX4 Wag	21.0	6	160	110
Datsun 710	22.8	4	108	93
Hornet 4 Drive	21.4	6	258	110
Hornet Sportabout	18.7	8	360	175
Valiant	18.1	6	225	105

Available themes: `"nejm"`, `"apa"`, `"nature"`, `"lancet"`, `"minimal"`.

The global theme set via `t2f_theme_set()` is respected by `zzt2f()` as well:

```
t2f_theme_set("nejm")
zzt2f(mtcars[1:6, 1:4], filename = "auto_themed")
```

Captions and Footnotes

```
fn <- t2f_footnote(
  general = "Data from Motor Trend US magazine (1974).",
  symbol = c("p < 0.05", "p < 0.01")
)

zzt2f(mtcars[1:6, 1:4],
      caption = "Motor Trend Car Road Tests",
      footnote = fn,
      filename = "captioned_table",
      sub_dir = output_dir,
      format = "png")
```

	mpg	cyl	disp	hp
Mazda RX4	21.0	6	160	110
Mazda RX4 Wag	21.0	6	160	110
Datsun 710	22.8	4	108	93
Hornet 4 Drive	21.4	6	258	110
Hornet Sportabout	18.7	8	360	175
Valiant	18.1	6	225	105

Data from Motor Trend US magazine (1974).

* p < 0.05

† p < 0.01

Table 1: Motor Trend Car Road Tests

Footnotes from `t2f_footnote()` are translated to plain-text notes below the table. General, numbered, alphabetic, and symbol notation styles are all supported.

Spanning Headers

Group columns under spanning headers using `t2f_header_above()`:

```
hdr <- t2f_header_above(
  " " = 2,
  "Performance" = 2,
  "Engine" = 1
)

zzt2f(mtcars[1:6, 1:4],
      header_above = hdr,
      filename = "header_table",
      sub_dir = output_dir,
      format = "png")
```

		Performance		Engine
	mpg	cyl	disp	hp
Mazda RX4	21.0	6	160	110
Mazda RX4 Wag	21.0	6	160	110
Datsun 710	22.8	4	108	93
Hornet 4 Drive	21.4	6	258	110
Hornet Sportabout	18.7	8	360	175
Valiant	18.1	6	225	105

Column Alignment

Alignment is auto-detected (numeric columns right-aligned, others left-aligned) or set explicitly:

```r
zzt2f(iris[1:6, ],
      align = c("r", "r", "r", "r", "l"),
      filename = "aligned_table",
      sub_dir = output_dir,
      format = "png")
```

Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
5.1	3.5	1.4	0.2	setosa
4.9	3.0	1.4	0.2	setosa
4.7	3.2	1.3	0.2	setosa
4.6	3.1	1.5	0.2	setosa
5.0	3.6	1.4	0.2	setosa
5.4	3.9	1.7	0.4	setosa

A single character applies to all columns:

```r
zzt2f(mtcars[1:6, 1:4], align = "c")
```

Row Stripe Color Override

Override the theme's stripe color with `scolor`:

```r
zzt2f(mtcars[1:6, 1:4],
      scolor = "blue!15",
      filename = "blue_stripes",
      sub_dir = output_dir,
      format = "png")
```

	mpg	cyl	disp	hp
Mazda RX4	21.0	6	160	110
Mazda RX4 Wag	21.0	6	160	110
Datsun 710	22.8	4	108	93
Hornet 4 Drive	21.4	6	258	110
Hornet Sportabout	18.7	8	360	175
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LaTeX color syntax (`"blue!15"`, `"gray!5"`) is automatically translated to hex. Hex values (`"#E6E6FF"`) pass through unchanged.

Working with Statistical Objects

`zzt2f()` operates on data frames only. For statistical objects, extract the data frame first, then pass it to `zzt2f()`:

```r
model <- lm(mpg ~ cyl + hp + wt, data = mtcars)
coef_df <- broom::tidy(model, conf.int = TRUE)
coef_df$p.value <- format_pvalue(coef_df$p.value)

zzt2f(coef_df,
      caption = "Linear Model Coefficients",
      theme = "nejm",
```

```
      filename = "lm_typst",
      sub_dir = output_dir,
      format = "png")
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	38.7517874	1.78686403	21.687038	<0.001	35.0915623	42.412012412
cyl	−0.9416168	0.55091638	−1.709183	0.098	−2.0701179	0.186884238
hp	−0.0180381	0.01187625	−1.518838	0.140	−0.0423655	0.006289293
wt	−3.1669731	0.74057588	−4.276365	<0.001	−4.6839740	−1.649972191

Table 1: Linear Model Coefficients

This two-step pattern (tidy then render) provides full control over column selection, rounding, and formatting before the table is generated.

Quick Reference

zzt2f() Parameters

Parameter	Description	Default
`filename`	Base name (no extension)	deparsed input
`sub_dir`	Output directory	`"analysis/figures"`
`format`	`"pdf"`, `"png"`, or `"svg"`	`"pdf"`
`dpi`	PNG resolution	300
`caption`	Table caption	NULL
`align`	Column alignment (l/c/r)	NULL (auto)
`theme`	Theme name or object	NULL
`scolor`	Row stripe color override	NULL
`footnote`	`t2f_footnote()` object	NULL
`header_above`	`t2f_header_above()` object	NULL
`verbose`	Print progress messages	FALSE

Parameters from t2f() Not Available in zzt2f()

These are LaTeX-specific and have no Typst equivalent:

- `extra_packages`, `document_class`
- `caption_short`, `label`
- `longtable`
- `crop`, `crop_margin`
- `collapse_rows`

Choosing Between t2f() and zzt2f()

	`t2f()`	`zzt2f()`
Backend	LaTeX (kableExtra)	Typst (tinytable)
Dependencies	pdflatex + pdfcrop + ImageMagick	Typst CLI (~40 MB)
Output formats	PDF (+ PNG/SVG via convert)	PDF, PNG, SVG natively
S3 dispatch	Yes (lm, glm, rms, …)	No (data frames only)
Cropping	Required (pdfcrop)	Not needed (auto-sized)
Decimal alignment	siunitx S columns	Not yet supported

	`t2f()`	`zzt2f()`
Multi-page tables	longtable	Not yet supported
Collapse rows	multirow	Not yet supported

Use `t2f()` when you need S3 dispatch, siunitx decimal alignment, or multi-page tables. Use `zzt2f()` when you want fast compilation with minimal system dependencies, or when you need native PNG/SVG output without ImageMagick.

System Requirements

- R >= 4.1 (for native pipe `|>`)
- tinytable R package
- Typst CLI (>= 0.11)

Check your setup:

```
check_typst_deps()
```